

Early axonal loss predicts long-term disability in chronic inflammatory demyelinating polyneuropathy



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HIGHLIGHTS

- Early axonal loss can accurately predict the long-term clinical outcome in patients treated for CIDP.
- Early axonal loss is associated with the long-term axonal loss in patients treated for CIDP.
- Delay in treatment is associated with worse axonal loss and clinical outcome.

ABSTRACT

Objective: To investigate early pre-treatment nerve fiber loss as a predictor of long-term clinical outcome in chronic inflammatory demyelinating polyneuropathy (CIDP).

Methods: In 14 patients, motor and sensory conduction studies of the median, fibular, and sural nerves were performed at pre-treatment and follow-up 11–28 years later. Z-scores of amplitudes were combined as biomarkers of axonal loss and Z-scores of conduction properties as demyelination scores. The axonal loss was further examined by electromyography (EMG) and motor unit number estimation. Axonal and demyelination scores were compared to clinical outcomes in the Inflammatory Rasch-built Overall Disability Scale, the Neuropathy Impairment Score, and dynamometry.

Results: At follow-up 12 patients walked independently, one needed support and one could not walk. The initial and follow-up axonal and demyelination scores were markedly abnormal. The initial axonal loss but not demyelination was strongly associated with both the follow-up axonal loss and the clinical measures. Moreover, delay of treatment initiation negatively influenced the axonal scores and clinical outcomes.

Conclusion: In this hypothesis generating limited study, we found that axonal loss at early CIDP was highly predictive for long-term nerve fiber loss and disability.

Significance: The study indicates that prompt initiation of treatment to prevent nerve fiber loss is necessary for outcome in CIDP.

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Abbreviations: ABP, abductor pollicis brevis; AT, anterior tibial; CIDP, chronic inflammatory demyelinating polyneuropathy; CMAP, compound muscle action potential; DML, distal motor latency; EDB, extensor digitorum brevis; EMG, electromyography; GBS, Guillain-Barré syndrome; I-RODS, Inflammatory Rasch-built Overall Disability Scale; IKS, isokinetic strength; MNCV, motor nerve conduction velocity; MUNE, motor unit number estimation; NCS, nerve conduction study; NIS, Neuropathy Impairment Score; SNAP, sensory nerve action potential; SNCV, sensory nerve conduction velocity.

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1. Introduction

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an immune-mediated polyneuropathy (McCombe et al., 1987; Prineas and McLeod, 1976) associated with a variable degree of axonal loss (Nagamatsu et al., 1999). Nerve conduction studies consistent with demyelination are mandatory for the diagnosis (Cornblath et al., 1991; Van den Bergh et al., 2010; Van den Bergh and Piéret, 2004), whereas the occurrence and the role of the axonal loss are less well-defined. Previously axonal loss was considered a secondary event in demyelinating disorders

(Bouchard et al., 1999; Said, 2006; Said et al., 1981), whereas recent studies indicate that axonal damage occurs early in the disease (Barnett et al., 2016). Several electrophysiological studies have shown axonal loss with denervation of muscle fibers at electromyography (EMG), reduced amplitude of the compound muscle action potential (CMAP), and a reduced number of motor units at motor unit number estimation (MUNE) (Kuwabara et al., 2006; Paramanathan et al., 2016; Sghirlanzoni et al., 2000).

Long-term CIDP is associated with a variable degree of disability, motor weakness with impaired walking performance being a frequent disturbance (Al-Zuhairy et al., 2020; Dyck et al., 1975; McCombe et al., 1987). As for the Guillain-Barré Syndrome (Willison et al., 2016), it is a general assumption that axonal loss in CIDP is the pathophysiological mechanism that underlies permanent disability (Bouchard et al., 1999; Mathey et al., 2015; Said et al., 1981). In a long-term CIDP study, disability was associated with the axonal loss but not with demyelination (Harbo et al., 2008a) whereas other studies were unable to establish such a connection (Kerasnoudis et al., 2015).

Due to its heterogeneous phenotype, it has been difficult to define biomarkers that may predict the course of CIDP and the degree of permanent disability. Early axonal loss in chronic CIDP could represent a long-term deficit, and we hypothesize that early axonal loss is a predictor of long-term axonal loss and long-term clinical disability. If the axonal loss is decisive for long-term disability the therapeutic strategy at the time of diagnosis might need a more comprehensive consideration in the future. To test this hypothesis, a study of the long-term disability and the electrophysiological changes was carried out in a group of patients 10–30 years after diagnosis. We examined whether the degree of axonal loss and *not* demyelination at the time of treatment initiation was predictive for, 1) the severity of axonal loss at follow-up, and 2) the long-term disability. As a biomarker of axonal loss, we calculated an *axonal score* from the deviations of the amplitudes of the compound muscle (CMAP) and sensory nerve (SNAP) action potentials from controls, and as a biomarker of demyelination, a *demyelination score* from the deviations of the motor and sensory conduction velocities and the decay of the CMAP between distal and proximal stimulation sites. As biomarkers of the clinical status, we used the Rasch-built Overall Disability Scale for immune-mediated peripheral neuropathies (I-RODS), the Neuropathy Impairment Score (NIS), and measurement of isokinetic strength (IKS).

2. Material and methods

2.1. Study population

The study population consisted of a subgroup of the 51 patients with probable or definite CIDP (Van den Bergh et al., 2010) treated with immunotherapy (i.e. immunoglobulins, plasma exchange, corticosteroids, azathioprine, or cyclophosphamide) during the period between Jan 1st, 1985 and Dec 31st, 2006 previously described (Al-Zuhairy et al., 2020). Exclusion criteria were other neuropathies, other neurological disorders, and disabling musculoskeletal disorders. A total of 16 eligible CIDP patients were identified in the Capital Region of Denmark in whom the pre-treatment and follow-up electrophysiological studies would be performed with the same methodology and comparable with the same control material. Two patients did not give consent to follow-up. The characteristics of the 14 participating patients (four females and 10 males) are presented in Tables 1 and 2. The time elapsed from symptom onset to the pre-treatment (initial) examination was 1.0 year (IQR: 0.0 – 3.0) and 17.5 years (IQR: 15.0 – 21.0) to the follow-up. The ages of the patients at the initial electrophysiological examination was 41.8 years (IQR: 33.0 – 54.0) and 61.0 years (IQR: 47.0 – 71.0) at the follow-up with an interval of 14.3 years

Table 1

Demographics and clinical data in 14 patients with chronic inflammatory demyelinating polyneuropathy (CIDP).

Variable	Median (Interquartile range)
Age (years)	61.0 (47.0 – 71.0)
Sex, M:F (n)	10:04
Age at onset (years)	41.0 (30.0 – 50.0)
Years since onset of CIDP (years)	17.5 (15.0 – 21.0)
Time since first contact to neurologist (years)	15.0 (14.0 – 21.0)
Duration of CIDP until treatment initiation (years)	1.0 (0.5 – 3.5)
The interval between initial and follow-up assessments (years)	14.3 (13.4 – 20.4)
Acute, GBS-like onset (n)	3
Walking status at initial visit (n)	
Walking independently	10
Walking with aids	0
No ambulation	4
Walking status at follow-up (n)	
Walking independently	12
Walking with aids	1
No ambulation	1
I-RODS score at follow-up (a.u.)	74.5 (60.0 – 83.0)
NIS score at follow-up (a.u.)	23.8 (21.0 – 30.5)
Isokinetic strength, normalized (%)	65.4 (50.5 – 79.7)
Current treatment (n)	
IgG	9
Azathioprine	1
No treatment	5
Combined axonal Z-score	
Initial	–3.5 (–7.2 – –2.3)
Follow-up ^a	–3.4 (–4.3 – –1.6)
Combined demyelination Z-score	
Initial	–4.9 (–8.2 – –3.0)
Follow-up ^b	–3.1 (–7.7 – –1.1)

M, male; F, female; GBS, Guillain-Barré syndrome; I-RODS, Inflammatory Rasch-built Overall Disability Scale; NIS, Neuropathy Impairment Score; IgG, Immunoglobulin G; a) $p = 0.2$; b) $p = 0.01$

(IQR: 13.4 – 20.4) between the studies. Three patients had an acute, GBS-like onset and the course was chronic or diphasic in all but one patient, who had a relapsing and remitting course for several years before the chronic phase. Nine patients still received immune-modulating therapy at follow-up.

2.2. Study design

The follow-up study included data on disease onset, course and duration combined with clinical and electrophysiological status at follow-up. To obtain the status at the follow-up study, all eligible patients were examined clinically including completion of the Inflammatory Rasch-built Overall Disability Scale (I-RODS), Neuropathy Impairment Score (NIS) and isokinetic strength (IKS) using dynamometry. They were interviewed on disease course and treatment and underwent an electrophysiological examination.

The Local Ethics Committee of the Capital Region (H-17017657) and The Danish Data Protection Agency (2012–58-0004) approved the protocol. All CIDP patients gave written informed consent according to the Declaration of Helsinki.

2.3. Clinical evaluation

Disability was evaluated with the Rasch-built Overall Disability Scale (R-ODS) for immune-mediated peripheral neuropathies (I-RODS) which is a self-reporting questionnaire evaluating limitations of physical activity and social participation with scores ranging from 0 (most severe disability) to 100 (no disability) following translation of the RAW sum score to a centile metric value (van Nes et al., 2011).

Neurological impairment was evaluated using the Neuropathy Impairment Score (NIS) (Dyck et al., 1997). Grading of muscle

Table 2

Individual clinical and electrophysiological data in 14 patients with chronic inflammatory demyelinating polyneuropathy (CIDP).

ID	Sex	Age (years)	Age at onset (years)	Duration of treatment (years)	Treatment status	I-RODS score (a.u.)	Electrophysiology follow-up (years)	Initial combined axonal Z-score	Current combined axonal Z-score	Initial combined demyelination Z-score	Current combined demyelination Z-score
1	M	69	58	12	Treatment	65	11.7	−2.3	−2.0	−4.2	−1.1
2	M	44	30	10	No treatment	60	14.3	−7.2	0.3	−9.9	−0.8
3	M	47	30	14	Treatment	73	14.1	−3.0	−2.5	−4.1	−4.3
4	M	71	46	22	No treatment	80	24.2	−2.9	−0.5	−1.7	0.2
5	F	48	34	14	Treatment	76	13.4	−5.1	−3.2	−10.1	−7.7
6	M	56	39	13	Treatment	100	14.2	−2.8	−4.3	−8.2	−7.9
7	F	51	22	18	Treatment	50	20.7	−9.5	−9.3	−7.8	−9.5
8	M	47	32	8	No treatment	100	13.4	−1.6	−4.0	−4.8	−3.0
9	M	72	54	12	Treatment	79	12.5	−4.0	−3.6	−6.3	−4.0
10	F	45	24	3	No treatment	69	20.4	−5.8	−0.8	−2.7	−0.4
11	M	70	49	16	Treatment	28	15.8	−11.1	−12.6	−11.8	−8.4
12	M	73	43	28	Treatment	88	28.0	−2.3	−1.6	−4.9	−1.9
13	F	66	50	11.5	Treatment	32	13.6	−7.7	−4.5	−3.0	−3.1
14	M	75	56	0.5	No treatment	83	17.3	−2.1	−3.6	−2.6	−2.8

I-RODS, Inflammatory Rasch-built Overall Disability Scale; M, male; F, female

strength of 21 pairs of muscle groups was scored as follows: Normal strength = 0; 75% of normal strength = 1; 50% = 2; 25% = 3; paralyzed = 4; five pairs of deep tendon reflexes and four modalities of sensation at the index finger and hallux at both sides are scored from 0 (normal) to 2 (absent), resulting in a total maximum impairment score of 220.

Dynamometry (Biodex System Pro 4/3[®]; Biodex Medical Systems Inc., Shirley, NY, USA) was used for evaluation of isokinetic strength (IKS) at the elbow, wrist, knee, and ankle at the weakest side of the patient. In the case of equal strength, the side selection was made at random as in the controls (Harbo et al., 2012). To weight all muscle groups equally, normalized strength was expressed as a ratio between the measured and predicted value, the latter being obtained from data of 178 healthy subjects reported from our laboratory (Harbo et al., 2012, 2008b).

2.4. Electrophysiological studies

2.4.1. Nerve conduction studies (NCS)

NCS were intended to be performed in the same nerves at the time of treatment initiation and the follow-up examination. In most patients, the median (motor and sensory fibers), fibular (motor and sensory fibers), and sural (sensory fibers) nerves were examined using near-nerve electrodes, and EMG was performed in the abductor pollicis brevis (APB) and the anterior tibial (AT) muscles. In the three patients who did not consent to studies with needle electrodes, the same nerves were examined using surface electrodes. For patients in whom both sides were examined initially, the most affected nerves were examined at follow-up. In one patient (#12) with motor conduction block in the median, ulnar and radial nerves, motor fibers of the lower extremities were not examined at the first study, and in another patient (#8) the tibial nerve rather than the fibular motor was examined at the first study.

The examinations were carried out under temperature control at 35–37 °C using an infrared heating lamp as conventionally used in our department. Motor fibers of the median nerve were examined by supramaximal stimulation of the nerve through surface or near-nerve needle electrodes at the wrist and elbow. The needle was placed close to the nerve by adjusting the position until a threshold of < 1 mA was attained (Buchthal and Rosenfalck, 1966). The compound muscle action potential (CMAP) was recorded from the APB (Dantec 20 Hz–10 kHz, Skovlunde, Den-

mark) through surface electrodes placed in a belly-tendon montage (Fig. 1). Motor fibers of the fibular nerve were stimulated through needle or surface electrodes at the fibular head (FH) and ankle and the CMAP was recorded through surface electrodes placed over the extensor digitorum brevis muscle (EDB) (Supplementary Fig. 1). In both the median and fibular nerves, the amplitudes of the CMAPs were measured peak-to-peak and the decay between wrist and elbow was calculated. The distal motor latencies (DML) were measured from the distal stimulus site to the first deviation of the CMAP from baseline and the motor nerve conduction velocities (MNCV) were calculated between proximal and distal stimulus sites.

Sensory fibers of the median nerve were examined by stimulating digit 1 and 3 and recording the compound sensory action potential (SNAP) at the near-nerve needle electrodes at wrist and elbow (Fig. 1) or by stimulating digit 2 and recording the SNAP at the surface electrode at the wrist (Dantec 200 Hz–4 kHz, Skovlunde, Denmark). The distal conduction velocities (distSNCV) from digit to wrist and proximally between the wrist and elbow (SNCV) were calculated. When using near-nerve needle electrodes at the fibular nerve, sensory fibers were studied by stimulating the superficial fibular nerve at the superior retinaculum at the ankle and recording the SNAP through the needle electrodes placed at FH (Supplementary Fig. 1). The sural nerve was studied between mid-calf and ankle through needle or surface electrodes and the SNAP was recorded through orthodromic (Supplementary Fig. 2) or antidromic stimulation. The amplitudes were measured peak-to-peak and the SNCV was calculated to the first positive phase of the SNAP. The SNAPs were averaged and when using near-nerve needle electrodes this allowed distinction of amplitudes of $\geq 0.1 \mu\text{V}$ from noise (Behse and Buchthal, 1978; Krarup, 2004, p. 2004; Krarup-Hansen et al., 2007).

2.4.2. Electromyography (EMG) and motor unit number estimation (MUNE)

To ascertain the extent of motor axon loss, EMG was performed in the APB and AT muscles using disposable concentric needle electrodes with a recording area of 0.07 mm^2 . The activity during rest was recorded to investigate denervation activity, during weak effort to analyze motor unit potentials (MUP), and during maximal effort to record the recruitment pattern (Dantec, 2 Hz–10 kHz). MUP analysis was carried out by recording the activity at 10 sites

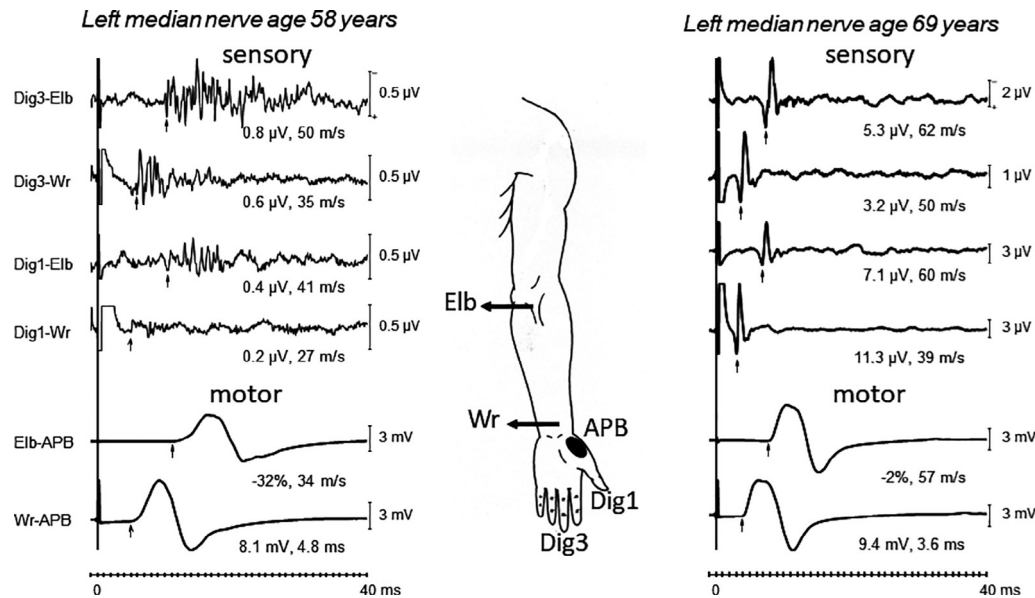


Fig. 1. Motor and sensory median nerve conduction studies from patient #1 at both examinations. Motor and sensory median nerve conduction studies from patient #1 (Table 2) at the time of diagnosis (left) and the follow-up study 11.7 years later (right). The amplitudes were measured peak-to-peak. The latencies were measured to the arrows. The motor conduction was carried out by stimulating the median nerve at elbow (Elb) and wrist (Wr) and recording the compound muscle action potential from the abductor pollicis brevis muscle (APB). The amplitude, distal motor latency, motor conduction velocity, and amplitude decay from wrist to elbow are indicated below the traces. The sensory nerve action potentials were recorded at Wr and Elb after stimulation at digit 1 (Dig1) and digit 3 (Dig3). The amplitudes of the sensory responses, the distal sensory conduction velocities, and the conduction velocities between Wr and Elb are indicated below the traces. Note that the reduced sensory and motor conduction velocities, the markedly reduced and dispersed sensory potentials, and the Elb/Wr ratio of the motor response recovered between the initial and the follow-up studies. The absolute amplitudes, latencies, conduction velocities, and motor response decay were converted to Z-scores to calculate the axonal and demyelination scores (see text and Table 3).

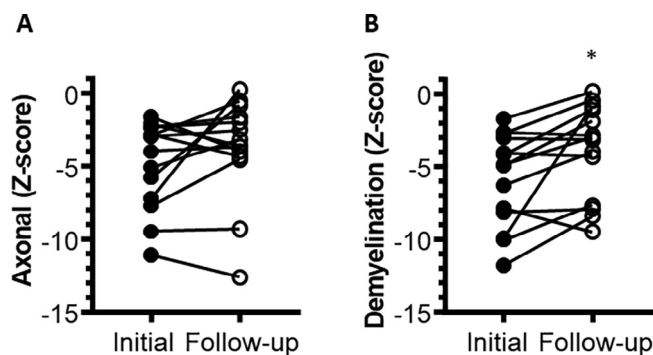


Fig. 2. Combined axonal and demyelination Z-scores from 14 patients with chronic inflammatory demyelinating polyneuropathy. The combined axonal (A) and demyelination (B) Z-scores from 14 patients with chronic inflammatory demyelinating polyneuropathy (CIDP) at the initial (filled points) and the follow-up studies (open circles). The axonal scores did not change from the initial to the follow-up studies ($p = 0.2$) whereas the demyelination scores improved (*, $p = 0.01$).

for 10 seconds and subsequent off-line decomposition using EMG-Tools (Nikolic and Krarup, 2011) and compared to our laboratory controls (Rosenfalck and Rosenfalck, 1975).

Axonal loss in the median nerve was quantified using MScanFit to count motor units in the APB. Surface electrodes were placed at the wrist and the CMAP was studied during increasing stimulus currents (Bostock, 2016). The findings in the patients were compared to MScanFit in 25 healthy subjects.

2.5. Statistical analysis

2.5.1. Electrophysiological analysis

Due to the long intervals between the initial and the follow-up studies, the deviations of electrophysiological parameters from control values took the influence of increasing ages on amplitudes,

conduction velocities, and EMG into account. The control values used for these calculations were derived from our laboratory controls (Horowitz and Krarup, 1992; Krarup and Trojaborg, 1996; Rosenfalck and Rosenfalck, 1975). The amplitudes of CMAPs and SNAPs decline exponentially with age (Horowitz and Krarup, 1992; Krarup and Trojaborg, 1996; Rosenfalck and Rosenfalck, 1975), and the recorded values from the CIDP patients were therefore logarithmically transformed.

Composite Z-values for axonal loss and demyelination were calculated. The values of the amplitudes of the CMAPs, the ratios of CMAP amplitudes at proximal/distal stimulation, the SNAP amplitudes, the DML, the MNCV, and the SNCV were converted to their corresponding Z-scores based on laboratory controls matched for age. For each patient, a combined axonal Z-score was calculated as the mean Z-score of the amplitudes of the median CMAP, fibular CMAP, and the mean amplitudes of the median, fibular and sural nerve SNAPs (higher values mean less axonal damage, Table 3). The combined demyelination Z-score (higher values mean less demyelination) was calculated based on conduction properties. In the median nerve the average Z-score of the inverse DML, the MNCV, and the proximal/distal amplitude ratio was calculated. For the fibular nerve, the average Z-score was based on the inverse DML and the MNCV. The mean Z-scores of the SNCVs of the median, fibular, and sural nerves were calculated (Table 3). Due to the more marked motor than sensory deficits in CIDP and the importance of motor conduction properties in the electrodiagnostic criteria of CIDP (Van den Bergh et al., 2010), we calculated combined Z-scores giving greater weight to the motor than sensory studies.

2.5.2. Applied statistics

Descriptive data were calculated using non-parametric statistics. Regression analyses were applied to assess associations. Comparisons of median values were performed using the Wilcoxon

Table 3

Calculations of combined axonal and demyelination Z-scores in Patient #1.

Combined Z-scores							
	Axonal 2006		Axonal 2018		Demyelination 2006		Demyelination 2018
APB CMAP ^{a)}	-1.308		-0.874		ABP inverse DML ^{h)}	-5.725	-1.336
EDB CMAP ^{b)}	-1.962		-3.685		Median MNCV ⁱ⁾	-6.127	-0.621
Median SNAP ^{c)}	-8.945	-3.634 ^{f)}	-2.023	-1.342	Median Wr/Elb ^{j)}	-3.241	0.141
Fibular SNAP ^{d)}	-0.622		-0.898		EDB inverse DML ^{k)}	-4.889	-2.444
Sural SNAP ^{e)}	-1.334		-1.106		Fibular MNCV ^{l)}	-5.033	-2.277
					Median SNCV ^{m)}	-4.935	-1.000
					Fibular SNCV ⁿ⁾	-3.357	-0.718
					Sural SNCV ^{o)}	0.541	0.661
Total Z-score ^{g)}	-2.301		-1.967			-4.192	-1.106

a) Amplitudes of the Compound Muscle Action Potential (CMAP) from the Abductor Pollicis Brevis (APB) and b) the Extensor Digitorum Brevis (EDB) muscles. c) Amplitudes of the compound Sensory Nerve Action Potential (SNAP) from the Median, d) Fibular, and e) Sural nerves. f) the average of the SNAP amplitude Z-scores. g) Average Z-score from the two motor nerves and the sensory score. h) Inverse Distal Motor Latency (DML) to the APB and i) the Motor Nerve Conduction Velocity (MNCV) from elbow to the wrist. j) The amplitude decay of the CMAP between wrist and elbow as related to distance. k) Inverse DML to the EDB, and l) the MNCV between fibular head and ankle. m) Sensory Nerve Conduction Velocity (SNCV) of the digit-to-wrist and the wrist-to-elbow segments of the median nerve, n) the ankle-to-fibular head of the fibular nerve, and o) the lateral malleolus-midcalf sural nerve.

Signed Rank Test and the Mann-Whitney *U* test for paired and non-paired comparisons respectively.

2.5.3. Power computation

The power computation was based on a linear regression model requiring an expected coefficient of determination, R^2 . No similar studies in CIDP have been conducted previously. In the present study, we expected the coefficient of determination, R^2 , to be relatively high. In the power computation, the R^2 was arbitrarily set to be 0.5 with a power of 0.8 and a 0.05 level of significance indicating that 11 patients were needed.

All statistics were performed using a SAS software package and Prism GraphPad.

Data availability

Anonymized participant data, study protocol, and statistical analysis plan can be obtained following contact with the corresponding author.

3. Results

3.1. Characteristics of the CIDP patients

The clinical data of the patients are presented in [tables 1 and 2](#). At the initial visit, four patients had no ambulation. In the follow-up study, 12 patients walked independently while one had lost ambulation and five of the patients were in remission.

3.2. Axonal loss as indicated by NCS, EMG, and MUNE

The amplitudes of the CMAPs and the SNAPs deviated strongly from normal controls and there was no difference between the combined axonal Z-scores at the initial and at the follow-up study ([Table 1](#), [Fig. 2A](#)). At univariate regression analysis, there was a significant association between axonal loss initially and at the follow-up study ($p = 0.01$, [Table 4](#)).

Table 4

Univariate regression analysis of the initial combined axonal Z-score vs. the follow-up combined axonal Z-score and the clinical measures.

Variable	Effect of a 1 a.u. increase in initial Z-score (95% CI)	R ²	p-value
Current combined Z-score	0.7 (0.2 – 1.3)	0.42	0.01
I-RODS	6.3 (4.0 – 8.6)	0.75	< 0.0001
NIS	-6.7 (-9.6 – -3.8)	0.68	0.003
Isokinetic strength	4.8 (2.2 – 7.4)	0.57	0.002

I-RODS, Inflammatory Rasch-built Overall Disability Scale; NIS, Neuropathy Impairment Score

At EMG of the AT, denervation activity was present in six of 12 muscles at the initial study but only in one of 11 muscles at the follow-up study. Fasciculations were present in 3–4 muscles at the initial and the follow-up studies. MUP analysis showed similar prolonged durations and increased amplitudes as indicated by the increased combined Z-scores, initially 2.2 (IQR: 0.8–5.4) and at follow-up, 3.1 (IQR: 1.8–4.4, $p = 0.9$). At the follow-up study, the APB showed mild-moderate denervation activity in eight of 11 patients. Eight patients also showed fasciculations, and in three they were profuse. MUP analysis showed prolonged durations and increased amplitudes as indicated by the Z-scores (4.8, IQR: 3.1–5.7). The increased Z-scores in the APB and the AT indicate collateral sprouting after partial denervation due to axonal degeneration. At MScanFit the number of motor units in the APB of CIDP patients was reduced by 35% ($p = 0.0003$) as compared to controls (data not shown).

3.3. Axonal loss as related to clinical findings

To study the predictive value of the axonal loss in early disease before treatment, the initial axonal Z-score was compared to the

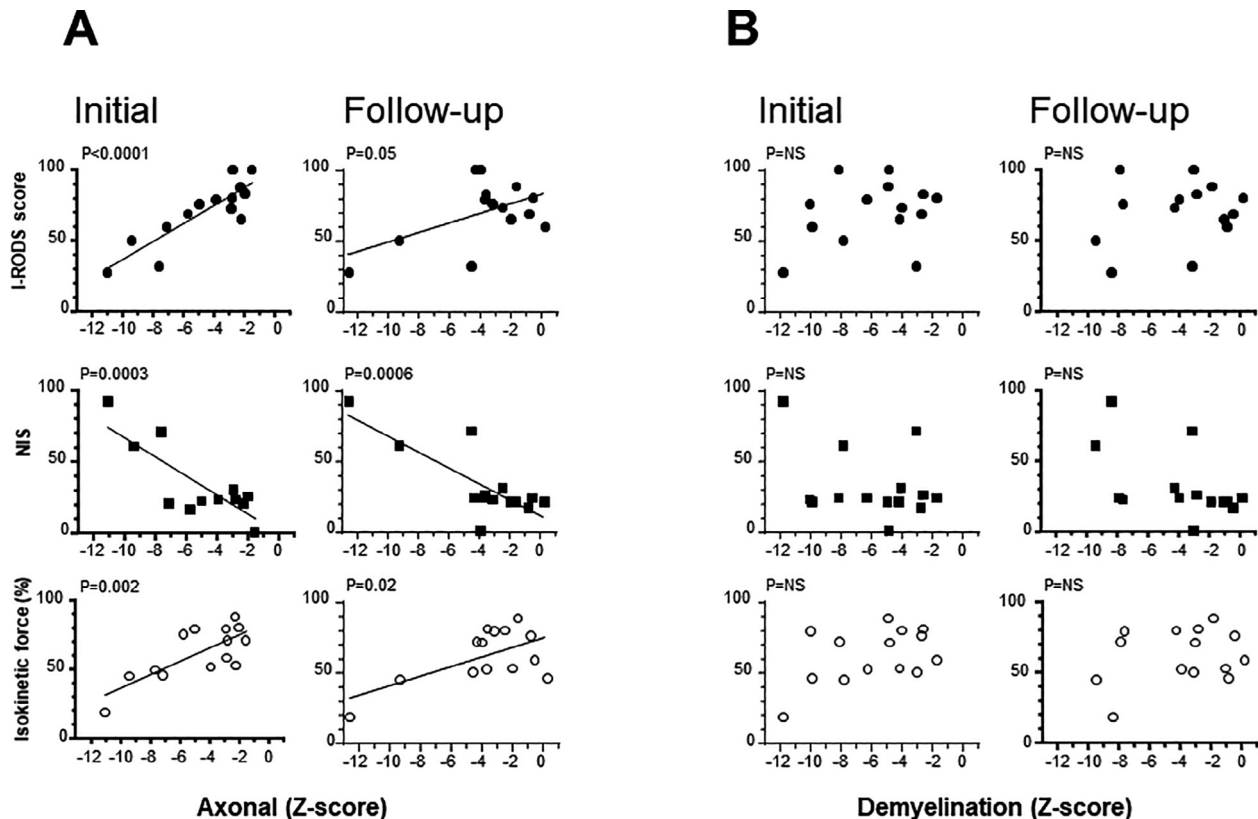


Fig. 3. Univariate regression analysis of the follow-up clinical scores versus the axonal and demyelination Z-scores at the initial and follow-up examinations. Univariate regression analysis of (A) the initial axonal and follow-up axonal Z-scores and (B) the initial and follow-up demyelination Z-scores and the follow-up Inflammatory Rasch-built Overall Disability Scale (I-RODS), the Neuropathy Impairment Score (NIS), and the isokinetic strength (%). At the initial axonal Z-scores, the associations were highly significant ($p < 0.0001$ – 0.002). At the follow-up axonal Z-scores, the association with the I-RODS score tended towards significance ($p = 0.05$) whereas the association with the NIS and isokinetic strength (%) were significant ($p = 0.0006$ – 0.02). In contrast, the relationships between the clinical scores and the demyelination Z-scores were not significant.

clinical measures represented by the I-RODS score, the NIS, and the normalized isokinetic strength at the follow-up study carried out 11–28 years after the initial study. As indicated in Table 4 and Fig. 3A, the initial axonal Z-score was highly predictive of the follow-up clinical disability and isokinetic strength ($R^2 = 0.57$ – 0.75 ; $p < 0.0001$ – 0.003). The predictive accuracy was further enhanced following the inclusion of age in the model, hence increasing the coefficient of determination (R^2) for the association between the initial axonal Z-score and the follow-up I-RODS for people of the same age to 0.80. The association between the follow-up axonal Z-score and the same clinical measures was weaker, the R^2 being 0.28–0.63 ($p = 0.0006$ – 0.05) (Fig. 3A).

3.4. Demyelination as indicated by conduction properties

In the initial study, the elbow/wrist CMAP amplitude ratio of the median nerve was reduced in nine and at the follow-up study in seven patients. The combined demyelination Z-score at the initial study was significantly more abnormal ($p = 0.01$) than at follow-up (Table 1, Fig. 2B). The demyelination scores at the first and the follow-up studies were strongly associated ($R^2 = 0.46$; $p = 0.007$, data not shown).

3.5. Demyelination as related to clinical findings

Neither the initial nor the follow-up demyelination Z-scores were related to the follow-up I-RODS score, NIS, or isokinetic strength (Fig. 3B).

3.6. The relation between axonal and demyelination scores

The demyelination and axonal scores were associated at the initial ($R^2 = 0.32$, $p = 0.03$) and the follow-up studies ($R^2 = 0.63$, $p = 0.0007$) (data not shown).

3.7. The effect of treatment on axonal loss and demyelination

In the follow-up study, nine of the 14 patients still received immune-modulating treatment, and there were no significant differences in axonal and clinical scores between the non-treated and the treated patients. The demyelination score, however, was 75% less abnormal in the non-treated than in the treated patients ($p = 0.01$).

There was a slight, negative association between the 0–11-year delay from symptom onset until treatment initiation and the initial axonal score ($p = 0.03$) and a strong negative association with the follow-up axonal score ($p = 0.005$). The delay in treatment was slightly associated with the follow-up demyelination score ($p = 0.02$) and negatively influenced the follow-up NIS ($p = 0.02$).

4. Discussion

The main and novel finding of this study was a strong association between the early axonal loss at the time of treatment initiation of CIDP and the degree of disability and impairment 11–28 years later. The ages of the patients varied greatly between 44 and 75 years, but the association between axonal loss and clinical disability was only slightly affected by age in a multivariate analy-

sis. In contrast, the degree of demyelination was not predictive of clinical disability. Recognizing that diminished amplitudes may be associated with distal conduction block due to demyelination, EMG was performed to confirm axonal loss and collateral sprouting.

The objective of the study was to assess whether early axonal loss was a predictor of long-term disability by directly comparing results of nerve conduction studies at the time of diagnosis and long-term follow up at more than 10 years using identical methodology, including temperature control at 35–37 °C, to allow calculation of Z-scores using our validated age-controlled normal material. These requirements limited the number of patients that could be recruited from the larger clinical study of treated Danish CIPD patients previously conducted (Al-Zuhairy et al., 2020), with the consequence that our study have hypothesis generating power, only. The fact that our results are highly significant despite the limited number of subjects indicates that the associations indeed are strong. Small sample studies are, however, more prone to produce biased and less reproducible results (Fraley and Vazire, 2014).

The association between axonal loss and clinical disability has been observed by studies of patients with a disease duration of 4–6 years (Bouchard et al., 1999; Sghirlanzoni et al., 2000) though the exact time of assessment in relation to the start of symptoms in these studies was less well-defined. In accordance with the findings in the present report, a long-term study in patients with a mean disease duration of 8.1 years established an association between late axonal loss as measured by the CMAP and SNAP amplitudes and clinical disability, whereas demyelination was not related to the clinical score (Harbo et al., 2008a). In our study, we also found a relation between axonal scores and clinical scores at follow-up though this association was weaker than the relation between initial axonal loss and late disability. In this context, it should be noticed that axonal loss as measured by the CMAP amplitude, may be masked by collateral sprouting and compensatory re-innervation as found in chronic neuropathies and motor neuron disease (Jakobsen et al., 2018, 2017; Kristensen et al., 2019). In our study, motor unit number estimation indicated motor axon loss in the median nerve, and there was strong evidence of chronic partial denervation and re-innervation at EMG which after long-term disease could limit the use of CMAP Z-scores as evidence of axonal loss, whereas this error might be less critical at early disease.

In our study, no association between the degree of demyelination and the clinical disability at early or late CIPD could be established, which is consistent with other demyelinating neuropathies such as Charcot-Marie-Tooth disease type 1 where the clinical disability is independent of demyelination but strongly related to the axonal loss (Buchthal and Behse, 1977). This discrepancy between the relationship between demyelinating scores and clinical disability raises the important question of the relation between demyelination and axonal loss in demyelinating neuropathies. It is often assumed that axonal loss in CIPD occurs as a consequence of demyelination (Bouchard et al., 1999; Köller et al., 2005; Said et al., 1981). Molecular mechanisms to explain this hypothesis include the release of chemokines, cytokines (e.g. tumor necrosis factor alpha), and toxic mediators (e.g. matrix metalloproteinases) by activated macrophages (Barnett et al., 2016; Beppu et al., 2015; Dalakas, 1999; Friese et al., 2014). These immunological processes could facilitate simultaneous myelin and axon destruction, the latter previously being attributed to a bystander effect (Madrid and Wiśniewski, 1977). Alternatively, the axonal loss could be an independent disease mechanism as in the axonal form of the Guillain-Barré Syndrome (GBS). The mechanism involved in such pathogenesis could involve an immune attack on axonal epitopes facilitated by peri axonal macrophages entering at the node of Ranvier (Griffin et al., 1996).

Reports on the effect of treatment on axonal loss are conflicting. Improvement of CMAP amplitudes following immune-modulating treatment was described in some studies (Bril et al., 2010; Dyck et al., 1994), whereas 20 treated patients who were followed for a median duration of 34 months showed no significant changes of the CMAP amplitude at baseline and follow-up though a subgroup with clinical deterioration showed amplitude reduction (Fisse et al., 2019). In accordance with this observation, we found the same degree of axonal loss as indicated by the Z-scores during the initial and follow-up studies. In contrast, the degree of demyelination was diminished after more than 10 years of CIPD, suggesting that demyelination was more reversible than axonal loss. Consistent with this finding the patients in remission had significantly less evidence of demyelination than those still on treatment as reported elsewhere (Chin et al., 2015). Hence, our observations support the notion that the electrophysiological assessment of demyelination can be of value for the decision of whether therapy should be continued (Katzberg et al., 2017).

A previous, clinical, retrospective study from which the present subjects were recruited found an association between delay in treatment initiation and long-term disability (Al-Zuhairy et al., 2020). In the present study, we found that the delay in initiating immune-modulating treatment was associated with the amount of axonal loss at both the initial and the follow-up study in addition to the association with the neurological impairment at follow-up. The association between the latency until treatment initiation and the long-term clinical disability and impairment was stronger in the previously published larger data group. The retrospective nature of the study can, however, be hypothesis generating, only, and there could be a bias in the observed associations (Al-Zuhairy et al., 2020). Considering the profound predictive effect of initial axonal loss on long-term disability, there is a need for a reliable biomarker that can predict the degree of permanent disability before treatment initiation. Our data suggest that long-term disability can be predicted with an accuracy of 75%. The accuracy is further improved to 80% following the inclusion of age into the model. In conclusion, our findings in this limited study material indicate that the degree of axonal loss at the time of the start of treatment is predictive for disability at follow-up 1–3 decades later. To confirm the hypothesis that early axonal loss is a robust biomarker of long-term disability a longitudinal, follow-up study of a larger cohort of newly diagnosed CIPD patients is warranted.

Author contributions

AA: Contributed to the design of the study, writing the protocol, coordination, clinical assessments, statistical analyses, and writing and revising the manuscript.

JJ: Contributed to the design of the study and writing and revising the manuscript.

CK: Contributed to the design of the study, electrophysiological examination, statistical analyses, and writing and revising the manuscript.

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Declaration of interest

AA: Declarations of interest: None.

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CK: Declarations of interest: None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clinph.2020.12.017>.

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